GRAINGUARD · DESIGN DECISION RECORD

GrainGuard concept selection

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/grainguard		
Source	docs/decisions/0004-concept-selection.md @ 5c553ac		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	Concept selection between options A, B and C of GGD-OPT-001; all within 10 %, sent to Amish without a pick

0004: Concept selection

- **Date:** 2026-10-24
- **Status:** proposed, awaiting Amish. No option is recommended: the three weighted scores are within 10 % (STANDARDS section 21). Option A stays the design of record until Amish decides.

Context

Kit 1.8 asks every repo for at least two concept options with different working principles, scored against the user requirements, with the current design of a repo designed before kit 1.8 as option A (STANDARDS section 21). GrainGuard reached TRL 3 on option A, a sensor cable down the bin axis with a solar controller on a mast. The kit 1.8 upgrade of 2026-10-24 added two options with different principles (GGD-OPT-001), modelled each at detail level D3 (kit 1.9, `cad/src/model.py`), checked their science (GGD-STM-001, rows B1 to C3) and scored all three against the seventeen user requirements of GGD-URS-001 plus cost, buildability, safety and maintenance.

Options considered

Option	Working principle	Score (of 245)	Estimated cost per bin	What it gives up
A, center sensor cable with EMC fan control	Six T and RH pods on a rope read the grain; plenum-EMC fan rule; solar controller on a mast	193	USD 410.50 (priced BOM)	Cost (USD 150.50 over the target), empty-bin installation, roof load during unloading
B, headspace sensor head with CO_2 watch	A head under the peak reads the air leaving the grain and its CO_2; wall-mounted solar controller	186	About USD 370 (estimate)	The temperature profile (PR-001) and per-level moisture (PR-002)
C, fan-side weather controller	Outside-air EMC and a cooling-front model; powered from the relay kit	192	About USD 170 (estimate)	Every grain measurement: PR-001, PR-002 and PR-006

Decision

None yet. The options are within 10 % (B is 3.6 % below A, C 0.5 % below A), so all three go to Amish without a pick, recorded as open item 10 in GGD-DEC-001.

For his decision: option A is the only option that meets PR-001, PR-002 and, near the center, PR-006, the requirements behind the pitch. Options B and C would change the pitch and what GrainGuard is, which is his decision in any case. Whatever he picks, two ideas from the runners-up are worth grafting onto option A: powering the controller from the relay kit's 24 V supply where the starter is near (from C; removes the panel, battery and charge controller, USD 76, and the battery's cold and hydrogen issues), and the headspace CO_2 head as a priced option on the same shackle (from B; widens heating detection beyond the center core).

Consequences

- If Amish keeps option A, nothing in the design changes; `concept_chosen` becomes A and the two grafts can be put to him as separate decisions.

- If he picks B or C, the pitch, the PRS (PR-001, PR-002 and PR-006 would be restated or withdrawn), the BOM, the build plan, the drawings and the calculation note are redone for that option at the next session; the D3 models of B and C in `cad/src/ggd_options.py` are the starting point.
- `project.yaml` keeps `concept_chosen: A (proposed)` until he decides.